

Education

2018–2022 **Vietnam National University Ho Chi Minh City - University of Science**
Bachelor's Degree in Computer Science

Work Experience

May 2022 – **Software Engineer (Distributed Systems), Zalo, HCMC**

- Present
- Built automated cross-DC replication for **1,000+ in-house DB clusters**, achieving **99.9% consistency** and enabling warm-standby DR.
 - Designed metrics ingestion pipeline at **1M samples/sec** with **10s anomaly detection**; architected Thanos long-term store, cutting query latency 40% and storage cost **3.5x**.
 - **Encryption**: added in-memory encryption to Memcached, securing PII for 70M+ MAU with <2% latency impact.
 - **Backup & Restore**: engineered daily **10TB+** backup flow with automated recoverability tests and snapshot drift metrics vs prod/previous.
 - **Hermetic Build**: created reproducible build pipeline from source to binary, deployed to **2,000+ DB nodes** with zero-drift releases.
 - **Stack**: C++, Java, Linux, Prometheus, Thanos, Grafana, Apache Thrift, Docker, Kubernetes.

May 2021 – **Software Engineer Intern, VNG Corporation**

- Nov 2021
- Developed new features and maintained a card-games called Truco for Argentinians on server side.
 - Maintained tool used for analysing user behavior based on given metric log.
 - Technologies: Java, Memcached, Python, Hadoop, Spark, MySQL.

Oct 2020 – **Backend Developer, Free Contest**

- Jun 2021
- Designed and developed APIs server.
 - Technologies: TypeScript, ExpressJS, MySQL.

Skills

- **Languages**: C/C++, Python, Java.
- **Technologies**: Linux, Docker, Prometheus, Thanos, Apache Thrift.
- **Core Technologies**: Distributed Systems, Data Structures/Algorithms, System Reliability Engineering.
- Highly experienced with **algorithm** and **data structures**.

Honours and Awards

2021 **Top 10 in AIOT Innovation Works, HCMC**

Worked with a team of 4 developing a proctoring system to help bring fairness in academics.

2020 **Fourth prize in AI Challenge of AISocha 2020, HCMC**

A 3-week challenge about counting the number of vehicles in each type of vehicle moving in different directions in Ho Chi Minh City.

2018 **Second Prize in ACM-ICPC Vietnam Southern Provincial Contest.**

Provincial competition among **over 300 teams**, each consists of 3 university students from Vietnam, solving algorithm and data structure problems using various programming languages such as C++ and Java.

2018 **Third Prize in Vietnamese National Olympiad in Informatics.**

Individual competition among high school students to solve data structures and algorithm problems using C++.

Projects

2018 **cowboycoder.tech**

A website sharing problem solution and code, as well as series of learning tips written or translated in Vietnamese.

Publications

2020 **SRGAN with Total Variation Loss in Face Super-Resolution**, *Hai Nguyen-Truong, Khoa N. A. Nguyen, San Cao*

Published in 2020 7th NAFOSTED Conference on Information and Computer Science (NICS)

2020 **HCMUS at MediaEval 2020: Ensembles of Temporal Deep Neural Networks for Table Tennis Strokes Classification Task**, *Hai Nguyen-Truong, San Cao, N. A. Khoa Nguyen, Bang-Dang Pham, Hieu Dao, Minh-Quan Le, Hoang-Phuc Nguyen-Dinh, Hai-Dang Nguyen and Minh-Triet Tran*

Published in The MediaEval 2020 Multimedia Evaluation Workshop